Frequency Mixer
(Using 2D Fourier Transform and Spatial Frequency Decomposition)

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1 Title

- **Frequency Mixer** (Using 2D Fourier Transform and Spatial Frequency Decomposition)

2 Objective

- To analyse the spatial frequency content of input images using 2D Discrete Fourier Transform (DFT).
- To understand how different frequency ranges contribute to perception.
- To fuse two images such that one contributes fine details (high frequency) and the other provides overall structure (low frequency).

3 Background

- The human visual system interprets images based on different spatial frequencies.
- High spatial frequencies include fine details, while low frequencies include overall shapes.
- Image processing (using the Fourier Transform) allows us to decompose images into these frequency components.
- This decomposition helps perform various operations on images, such as:
  - Blurring
  - Edge detection
  - Image fusion
- One example of this concept is the Einstein–Marilyn illusion.

4 Problem Statement

- Given Two Grayscale Images
- Objectives:
  - Write the equations governing the 2D discrete variable Fourier transform.
  - Compute the 2D DFT of any one of the input images.
  - Plot its magnitude spectrum in both linear and dB scale.
  - Shift the spectrum so that low frequencies are centred.
  - Rotate the image 90 degrees anticlockwise and compare both spectrums.
  - Create a frequency mixer that mixes:
    - High-frequency content of one image with
    - Low-frequency content of the other image.

- Explain the results using 2D magnitude transfer functions.

5 Theory

1. Start with a 2D signal $I(x, y)$ of size $M \times N$, where:
   - $x = 0, 1, \ldots, M-1$ is the row index
   - $y = 0, 1, \ldots, N-1$ is the column index
2. Apply 1D DFT along each row (i.e., along the x-direction), for every fixed y:

$$G(u, y) = \sum_{x=0}^{M-1} I(x, y) \cdot e^{-j2\pi \frac{ux}{M}}$$

3. Apply 1D DFT along each column of the intermediate result $G(u, y)$ (i.e., along the y-direction), for every fixed u:

$$F(u, v) = \sum_{y=0}^{N-1} G(u, y) \cdot e^{-j2\pi \frac{vy}{N}}$$

4. Combine the two steps to get the final 2D DFT expression:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I(x, y) \cdot e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

5. To construct image from DFT , we use inverse 2D Discrete fourier transforms:

$$I(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) \cdot e^{j2\pi \left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Where:

- $I(x, y)$: Input image in the spatial domain.
- $F(u, v)$: Output image in the frequency domain (Fourier spectrum).
- x, y: Spatial coordinates (pixel positions) in the input image.
- u, v : Frequency domain coordinates.
- M, N : Dimensions of the image along x and y axes respectively.
- j : Imaginary unit, where $j^2 = -1$.
- $e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N}\right)}$: Complex exponential kernel used in the Fourier transform.

To shift the magnitude spectrum of DFT so that low frequencies are centered , we use np.fft.fftshift() function

$$\text{Here} \quad f_{\text{shifted}}(u,v) = f\left(u - \frac{M}{2},\ v - \frac{N}{2}\right)$$

property is used.

Also, we are asked to rotate the image 90 degrees anticlockwise and compare the spectra. therefore we use the rotation property, which says that if image I(x,y) is rotated by θ then DFT F(u,v) is also rotated by θ.

Now while fusing images, we are using low-pass and high pass filtered images in the frequency domain.

Here we can see that, linearity property of DFT is used

$$F\left\{a \cdot I_1(x,y) + b \cdot I_2(x,y)\right\} = a \cdot f_1(u,v) + b \cdot f_2(u,v)$$

Also to get a filtered image , we use the property that multiplication in the frequency domain corresponds to convolution in the spatial domain.

$$I_{\text{filtered}}(x,y) = \mathcal{F}^{-1}\left\{F(u,v) \cdot H(u,v)\right\}$$

All these properties are used for finding various tasks asked in questions , with the help of inbuilt functions of python libraries.

6 Tools and Techniques used

- Various inbuilt Python functions such as `np.fft.fft2`, `np.fft.fftshift`, etc., are used to calculate the various plots asked in the question.
- **Computation of the 2D Fourier Transform is done using the `np.fft.fft2()` function.**

6.1 Plotting magnitude spectra of cat input image in normal and dB form:

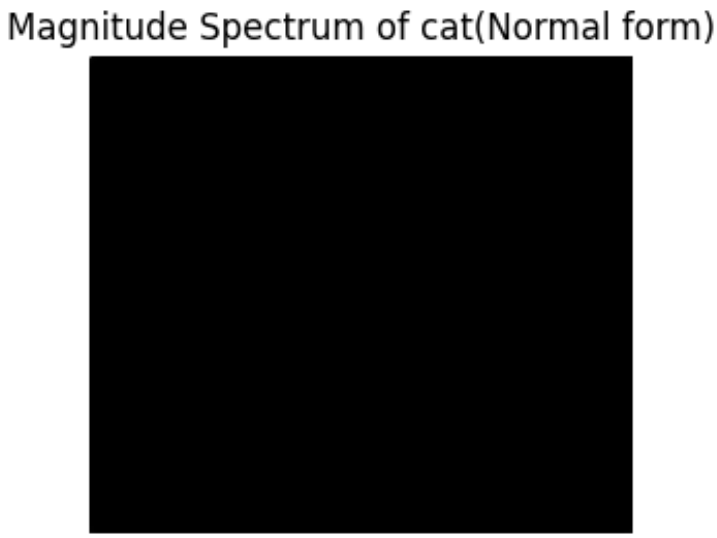

Figure 1: Magnitude Spectrum of cat (Normal Form)

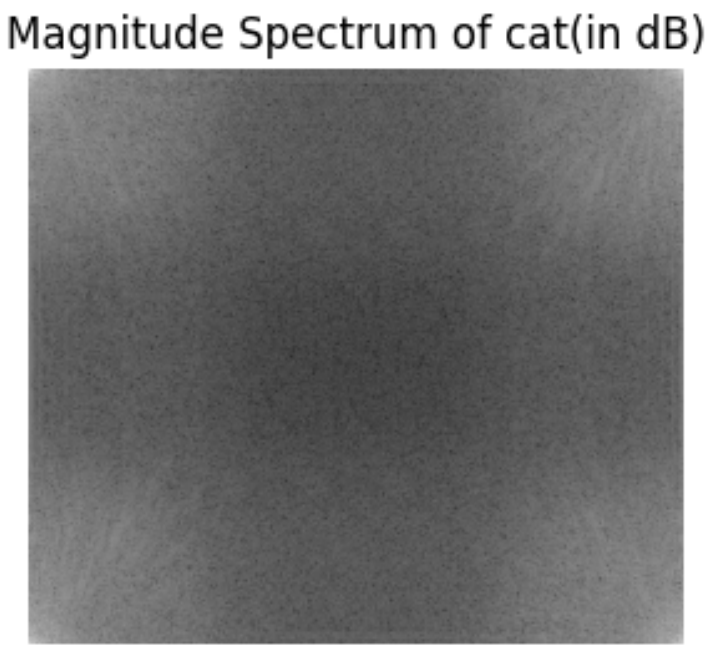

Figure 2: Magnitude Spectrum of cat (in dB)

- Here, we can see that no conclusion can be drawn from the normal form of the magnitude spectrum

- Hence, the dB (logarithmic) form is widely used for better visibility of details.
- From the dB form, we can conclude that:
  - **High frequencies are located at the centre of the spectrum.**
  - **Low frequencies lie at the corners.**

6.2 Shifting low frequencies at the centre of the magnitude spectrum

- To shift the low frequencies to the centre of the magnitude spectrum, the `np.fft.fftshift()` function is applied.
- This gives the magnitude spectrum of the shifted Fourier transform from the cat input image given.
- It is visible that low frequencies carry more energy as the image has smoother variations. Therefore, the brighter part in the dB plot corresponds to low frequencies.
- Now in the dB plot, we can see that the centre part is brighter than the corners and edges. Hence, **the centre of this plot is low frequencies, and the corners are high frequencies.**

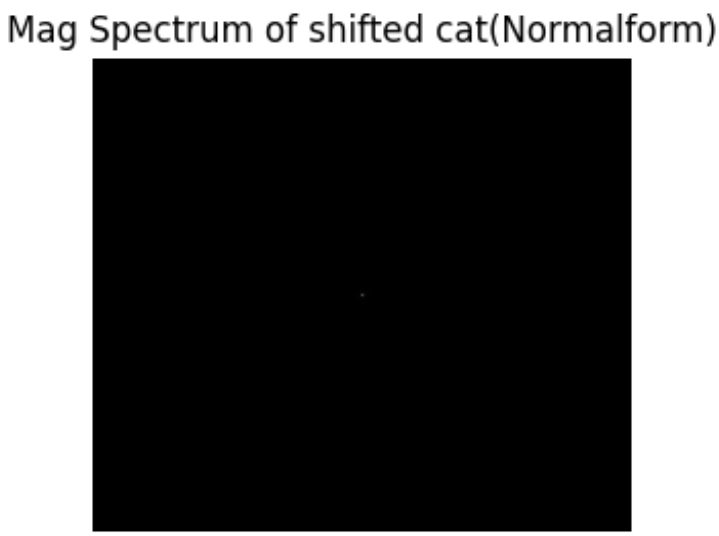

Figure 3: Magnitude Spectrum of shifted cat (Normal Form)

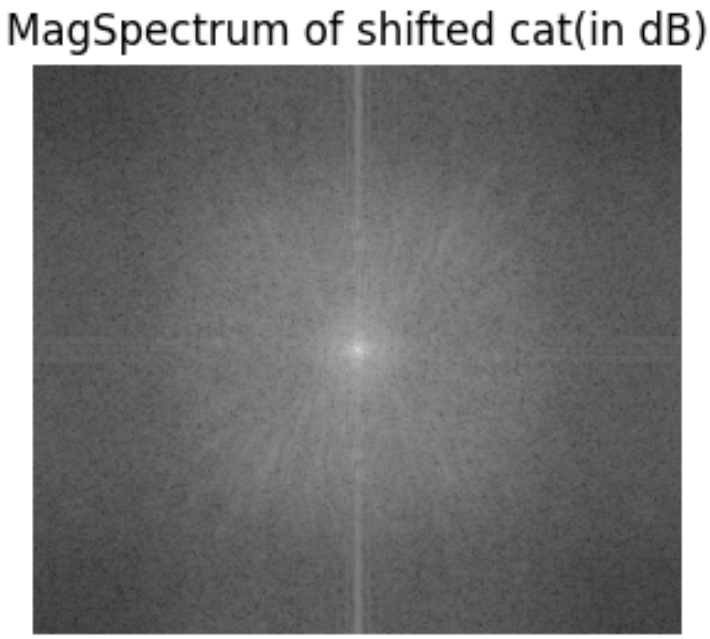

Figure 4: Magnitude Spectrum of shifted cat (in dB)

6.3 Rotating image anticlockwise

- It is done using `np.rot90()` function.

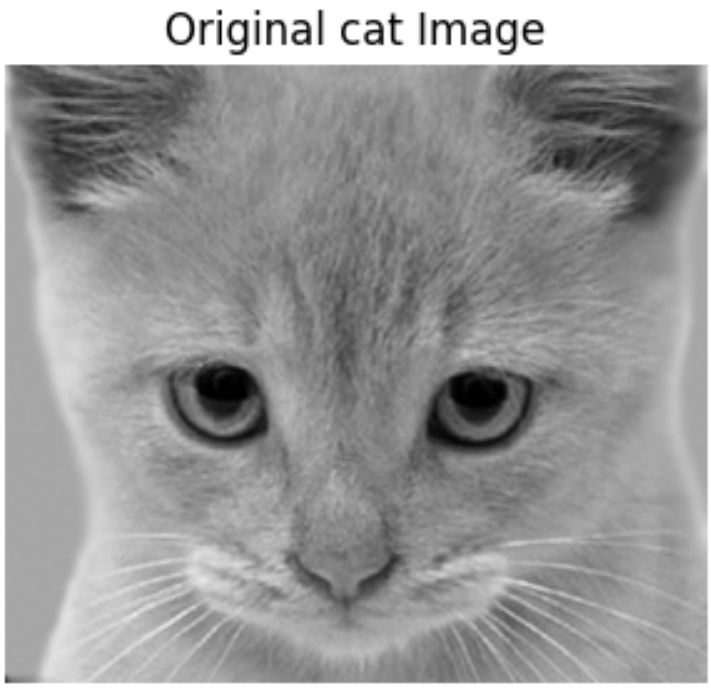

Figure 5: Original Cat Image

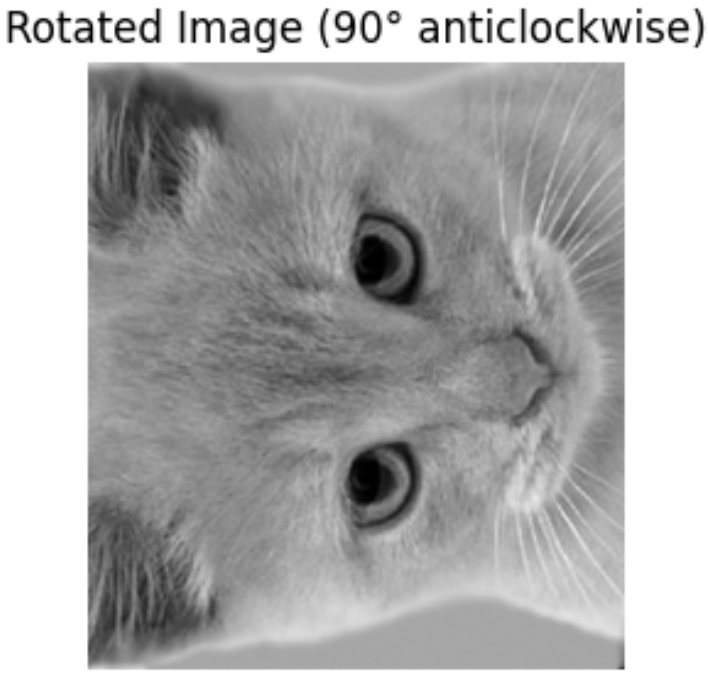

Figure 6: Rotated Image (90 degree anticlockwise)

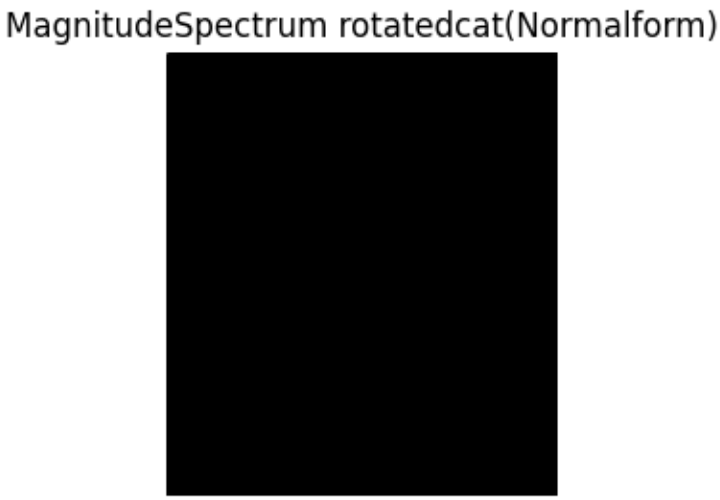

Figure 7: Magnitude Spectrum - rotatedCat (Normal Form)

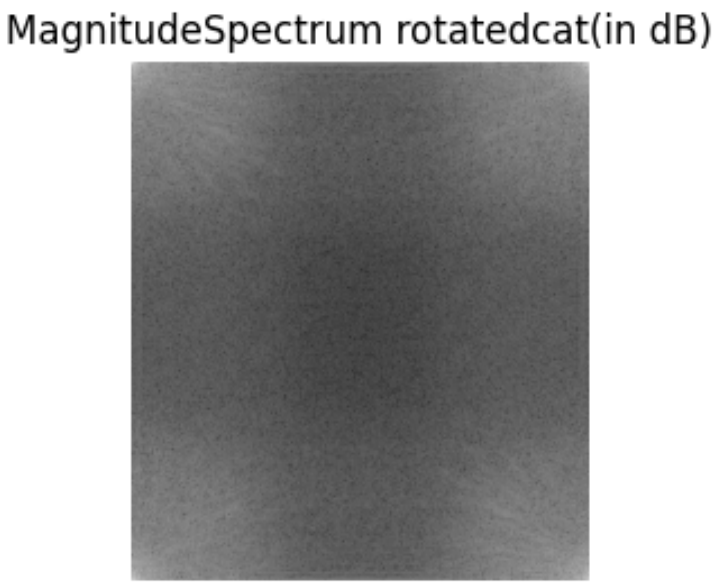

Figure 8: Magnitude Spectrum - rotatedCat (in dB)

- No conclusion can be drawn from it.
- Therefore the shifted fourier transform of the rotated image is considered.
- **Magnitude spectrum of shifted fourier transform of rotated image**

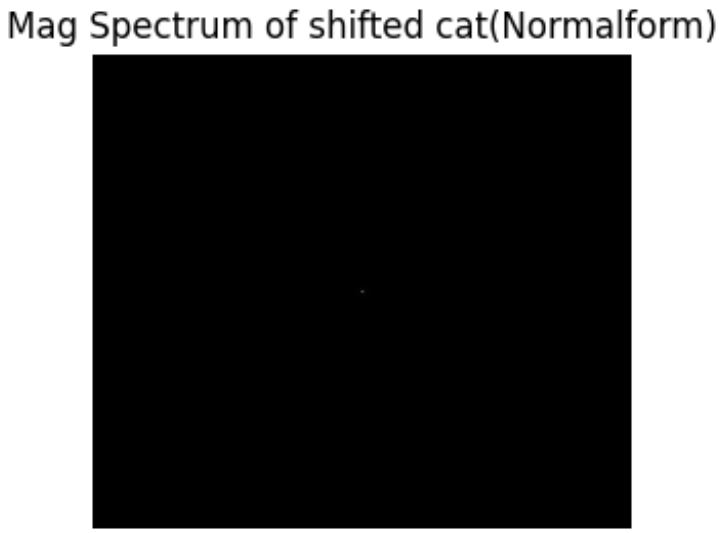

Figure 9: Magnitude Spectrum - rotatedShiftedCat (Normal Form)

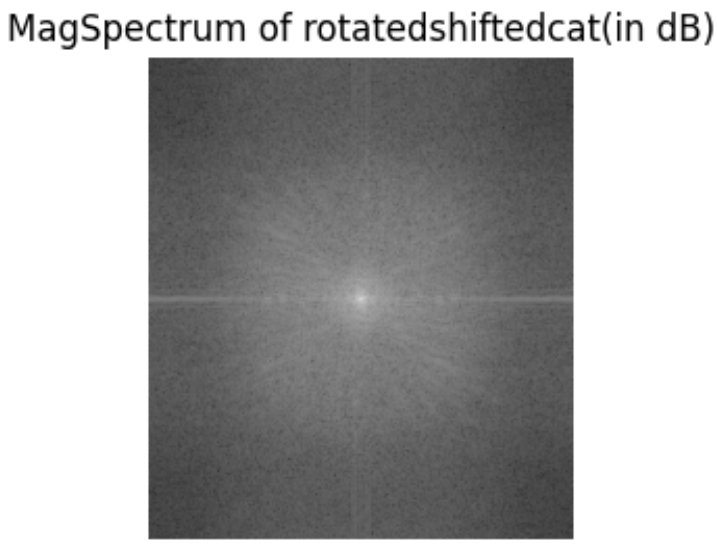

Figure 10: Magnitude Spectrum - rotatedShiftedCat (in dB)

- We can compare this dB form with the earlier dB form.
- **It is visible that the spectrum is also rotated 90 degrees like the image.**
- **The result is consistent with the property mentioned in theory.**
- Functions for low-pass marks and high-pass marks are used to get a fused image.

7 Method used for fusing two input images

- The Fourier Transform of both images is calculated to convert them to the frequency domain.
- Using a shifting function, the spectra are shifted so that low-frequency components are centred, which makes it easier to apply filters.
- A circular low-pass mask is created to extract the low-frequency components of the cat image. These frequencies correspond to:
  - Smooth intensity variations
  - The general structure of the image
- A complementary high-pass mask is created by subtracting 1 from the low-pass mask to obtain the high-frequency components of the dog image. These frequencies correspond to:
  - Fine details
  - Edges of the image
- These masks are then applied to the respective cat and dog images.
- The filtered frequency spectra (low from the cat and high from the dog) are added together to obtain the fused frequency spectrum.
- Finally, the inverse Fourier Transform is computed to get back the fused image.
- The resulting image contains:
  - Structural information from the cat image
  - Fine details and textures contributed by the dog image

8 Final Output

8.1 Fused image formed:

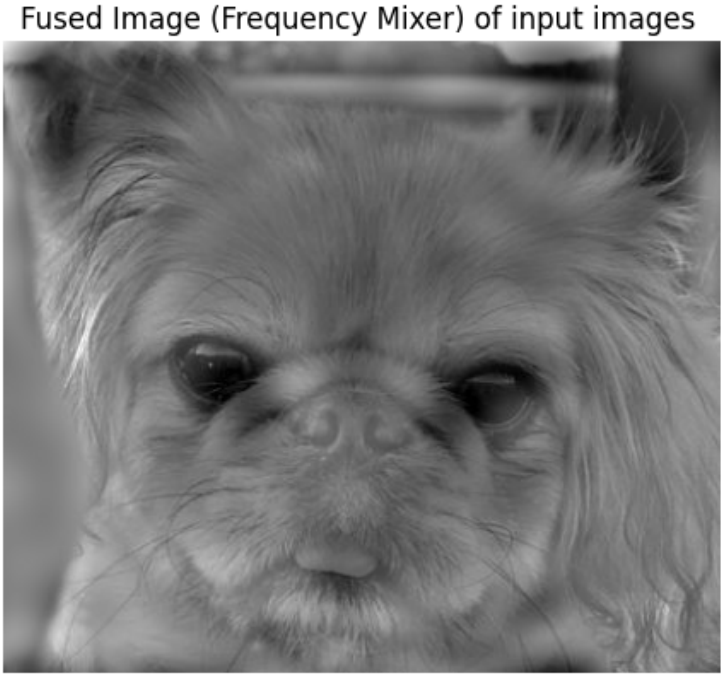

Figure 11: Fused Image (Frequency Mixer) of input images

8.2 Transfer function 2D plots:

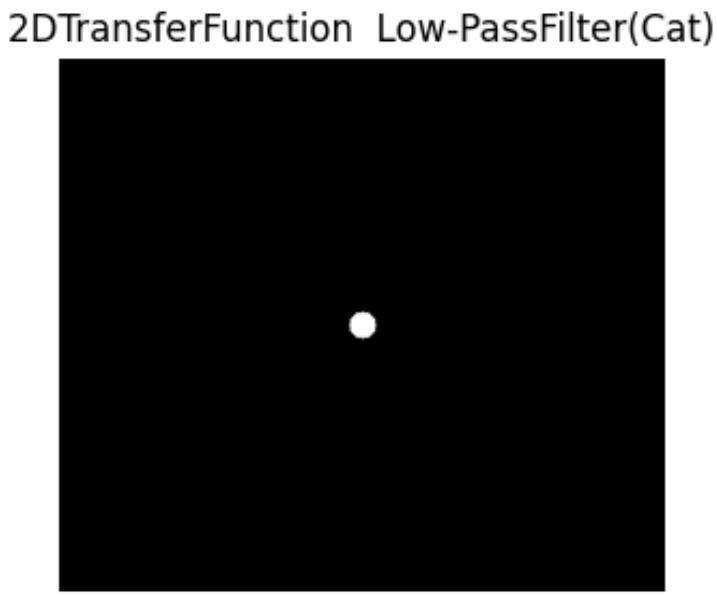

Figure 12: 2DTransferFunction Low-Pass Filter (cat)

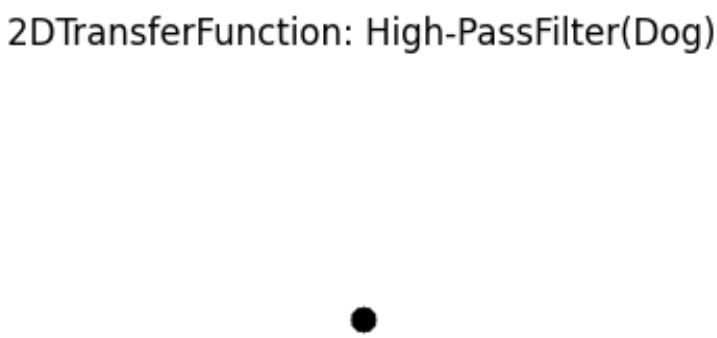

Figure 13: 2DTransferFunction High-Pass Filter (dog)

- From this, we can conclude that **there is no overlapping of frequencies, and no frequency component is lost from either of the images.**

9 Results

- The shifted spectrum has low frequencies positioned at the center.
- After a 90-degree rotation of the image, the corresponding spectrum also rotates by the same amount.
- The fused image successfully combines:
  - Fine details from one image
  - General structure from the other image
- Transfer function plots clearly show that:
  - A high-pass mask is used to extract fine details.
  - A low-pass mask is used to retain overall structure.

10 Conclusion

- The practical use of filters, the Fourier transform, and various properties of it can be seen in the question.